

Oxygen consumption of largemouth bass, *Micropterus salmoides*, in relation to swimming speed and temperature

F. W. H. BEAMISH

Department of Zoology, University of Guelph, Guelph, Ontario

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Oxygen consumption was determined for largemouth bass, *Micropterus salmoides*, of various sizes in relation to sustained swimming speeds. The logarithm of oxygen consumption for a given swimming speed and temperature increased linearly with the logarithm of weight. Slopes of regressions were considerably less than unity. In general, slopes for different swimming speeds at a given temperature did not differ significantly. The relation between logarithm of oxygen consumption and swimming speed is expressed as a series of parallel, linear regressions for the temperatures followed. For a given swimming speed, oxygen consumption increased with temperature from 10 to 34 C, the highest temperature followed.

Active oxygen consumption increased linearly with weight when expressed on a logarithmic grid. At each temperature the slope of the regression approached unity. The logarithm of standard oxygen consumption increased linearly with temperature from 10 to 34 C.

The logarithm of maximum sustained swimming speed increased linearly with total length at each temperature. For a given total length, maximum sustained swimming speed increased from 10 to 30 C. Between 30 and 34 C maximum speed decreased.

Introduction

Customarily, energy dissipated through locomotion is expressed as oxygen consumption. Metabolic expenditure is termed routine when swimming is spontaneous (Fry 1957) and sustained (forced long-term) or forced short-term when fish are induced to swim (Fry 1967). Metabolism at sustained swimming speeds excludes expenditures associated with excitability and hence is usually a more useful measurement than forced short-term.

This report describes the oxygen consumption of largemouth bass, *Micropterus salmoides*, of different size in relation to temperature and sustained swimming speeds and represents one part of a total study on the energetics of this species.

Materials and Methods

Largemouth bass were captured by seine from a small lake near Shakespeare, Ontario, and transported to our laboratory where they were held in 54-l tanks. Fish were kept for several months at 20 C, which was comparable to the temperature at time of capture. Analyses were made twice weekly for ammonia nitrogen, oxygen, and carbon dioxide (American Public Health Association 1960). Ammonia nitrogen generally was below 0.1 mg/l. Oxygen and carbon dioxide changed little from 8 and 1 mg/l respectively. Fish were fed to satiation once daily on a diet of beef liver and occasionally Atlantic herring, *Clupea harengus*, and lake emerald shiner, *Notropis atherinoides*. Photoperiod was maintained at 16 h throughout the study. Overhead incandescent lights provided an intensity of about 60 lux at the surface of the uncovered portion of holding tanks. Light intensity was gradually increased or decreased for $\frac{1}{2}$ h at either end of the photoperiod.

Before fish were used in an experiment they were acclimated to a given temperature by a daily change of one degree and allowed 1 month at the desired temperature. Food was withheld for 48 h before the start of an experiment (Beamish 1964b).

The respiration chamber was designed as a recirculating water tunnel and is described in Farmer and Beamish (1969). Briefly, water is driven by a centrifugal pump over a temperature-compensated oxygen electrode into an expansion chamber in which are located a series of screens or flow "straighteners," through a reduction cone, and into the portion of the chamber in which fish are forced to swim. Velocity profile in the swimming chamber was essentially constant. Two respiration chambers were used. Fish of 40 to 100 g were used in the smaller chamber which has a volume of 37 l. The length of the tunnel portion available to a fish is 36 cm with an inner diameter of 9.6 cm. This allowed space for free swimming without sacrificing sensitivity of oxygen change. The larger chamber has a volume of 168 l. The length of the tunnel portion available for swimming is 55 cm with an inner diameter of 19.5 cm. Fish of 150 to 400 g were used in the larger chamber.

The oxygen electrode in the respirometer was standardized regularly by the sodium azide modification of the Winkler method (American Public Health Association 1969). For a given oxygen concentration in the respiration chamber, the electrode registered constant readings at velocities of 30 cm/s and above. Below 30 cm/s electrode sensitivity was reduced. At the lowest velocity followed, 20 cm/s, the oxygen electrode registered less than that at the higher velocities for a given concentration of oxygen. Hence oxygen consumption was determined by chemical analyses at the lowest velocity.

The experimental procedure followed was to put a fish into a respirometer and gradually increase velocity to 30 cm/s. At most temperatures 30 cm/s was the slowest velocity at which largemouth bass swam continuously. However, at 10 and 15 C, smaller fish were exercised

initially at 20 cm/s. Oxygen concentration and temperature were recorded at 15-min intervals. Fish were exercised at the initial velocity until oxygen consumption remained about the same for three consecutive 15-min periods. Once a constancy of oxygen consumption was established water in the respirometer was exchanged with water saturated with oxygen. Thereafter water velocity was increased in increments of 10 cm/s, each for 30 min. The stepwise increment was repeated until fish fatigued. A blank was run for each experiment to determine oxygen consumption not attributable to fish. Oxygen consumption was then adjusted for each swimming speed. The procedure permitted calculation of maximum sustained swimming speed for a 30-min period in the manner described by Brett (1964). Extrapolation of the relation between oxygen consumption and swimming speed to the maximum sustained speed provided an estimate of active oxygen consumption.

Each fish was trained before a final experiment. The procedure was to exercise fish as outlined above once every third day for three trials. Generally results of the

third trial were accepted. The influence of training on oxygen consumption and maximum sustained swimming speeds is summarized in Fig. 1. At swimming speeds of 30, 40, and 50 cm/s, oxygen consumption decreased from the first to second trial. Between the second and sixth, an obvious change in oxygen consumption was not demonstrable. During the first trial at 60 cm/s, fish were unable to swim long enough to estimate oxygen consumption. In subsequent trials, swimming capacity increased and oxygen consumption was determined but for a period less than the regular 30 min. From the second to third trial, oxygen consumption decreased; thereafter little change was found in oxygen consumption. Similarly, maximum sustained swimming speed increased progressively from the first to third trial and thereafter remained about the same.

During an experiment, oxygen level in the respirometer was not allowed to fall below 70% of air saturation. Oxygen consumption of largemouth bass at this level did not differ appreciably from that when oxygen was above air saturation, irrespective of swimming speed.

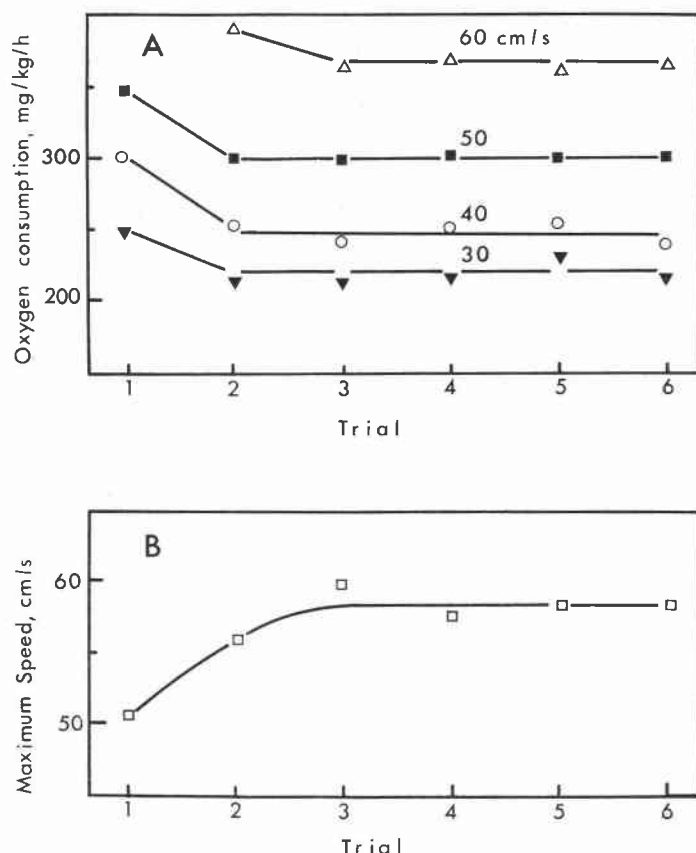


FIG. 1. (A) Oxygen consumption and (B) maximum sustained swimming speed of seven largemouth bass in relation to training trials. Oxygen consumption and maximum speed are expressed for a fish of 150 g. At 60 cm/s oxygen consumption is based on an interval of about 25 min as fish of 150 g were unable to swim for the regular 30-min period. Temperature was 25 C.

Four largemouth bass were subjected to the training procedure outlined above. Oxygen concentration during the first three trials averaged 80% of air saturation. For the fourth trial, oxygen concentration averaged 150% of air saturation. Significant differences in oxygen consumption for a given swimming speed were not observed at the two oxygen concentrations (Fig. 2). Moreover, the higher concentration of oxygen was of no advantage to the maximum sustained swimming speed.

Standard or resting metabolism was estimated by extrapolation of the relation between oxygen consumption and swimming speed to zero activity.

Linear regression lines were fitted according to the principle of least squares (Alder and Roessler 1964). Correlation coefficients (Alder and Roessler 1964) and 95% confidence intervals of slopes (Steel and Torrie 1960) were calculated for regression lines. Standard errors and deviations were calculated as outlined in Steel and Torrie (1960).

Results

The logarithm of oxygen consumption increased linearly with the logarithm of weight for a given swimming speed and temperature. However, it was not possible to determine the relation for all temperatures and swimming speeds (Table I). Over the thermal range tested, 5 to 34 C, fish did not swim long enough at 5 C at any speed irrespective of size to establish a reliable estimate of oxygen consumption. At other temperatures, small fish generally were unable to swim at the higher speeds. When the weight range for which oxygen consumption could be determined was appreciably reduced, a regression equation was not calculated. Similarly, regressions were not

calculated at 20 cm/s, as large fish swam poorly at this speed.

Slopes of regression equations in all cases were less than unity. In general, slopes for different swimming speeds at a given temperature did not differ significantly at $P < 0.05$. The one exception occurred at 30 C when the slope at 40 cm/s, 0.5580, was significantly different ($P > 0.05$) from that at 30, 0.6593, and 50 cm/s, 0.6217. For a given swimming speed an obvious pattern of change in slope was not demonstrated among the acclimation temperatures.

Oxygen consumption was calculated for a 150-g fish, the approximate overall mean weight, at each temperature and swimming speed from the appropriate regression equations in Table I. Where a regression equation was not available, a slope of 0.65 was used to adjust rates of oxygen consumption. The relation between logarithm of oxygen consumption and swimming speed is expressed for the various temperatures as a series of parallel linear regressions using the mean slope, 0.0085 (Fig. 3). For a given swimming speed, oxygen consumption increased from 10 to 34 C.

Active oxygen consumption increased linearly with weight for each temperature when expressed on a logarithmic grid. Equations for the regressions are as follows.

$$10\text{ C } \log Y = -0.6681 + 0.9725 \log X$$

$$15\text{ C } \log Y = -0.5590 + 0.9855 \log X$$

$$20\text{ C } \log Y = -0.0417 + 0.8068 \log X$$

$$25\text{ C } \log Y = -0.2511 + 0.9190 \log X$$

$$30\text{ C } \log Y = -0.0143 + 0.8836 \log X$$

$$34\text{ C } \log Y = -0.1336 + 0.9358 \log X$$

A more complete summary is given in Appendix I available from the Depository of the National Research Council of Canada* in Ottawa, Canada.

The logarithm of active oxygen consumption for a 150-g fish increased linearly with the logarithm of temperature from 10 to 30 C (Fig. 5). Between 30 and 34 C active oxygen consumption decreased. Over the linear phase the Q_{10} was 1.65. Standard oxygen consumption (zero swimming speed) increased linearly with temperature from

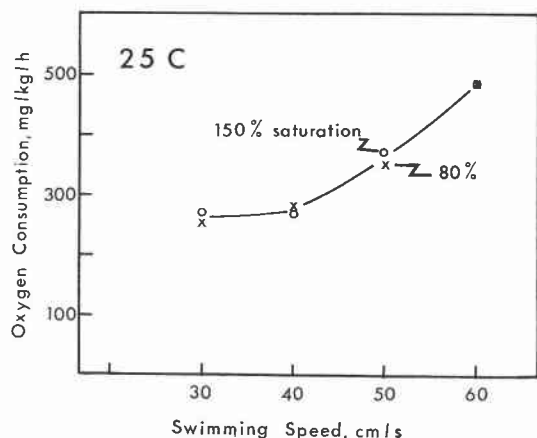


FIG. 2. Influence of oxygen concentration on oxygen consumption for a given swimming speed. Oxygen consumption is expressed as mg/kg h for the mean weight of the four fish used 43 g.

*The Depository of Unpublished Data, National Science Library, National Research Council of Canada, Ottawa, Canada.

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TABLE I
The relation between weight (X in g) and oxygen consumption (Y in mg/h) at different swimming speeds and temperatures

Acclimation temp., C	Swimming speed, cm/s	No. fish	Total length, cm		Weight, g		Equation	Correlation coefficient	Confidence interval of slope (95%)
			Mean	SD	Mean	SD			
10	20	8	18.6	1.4	89.4	22.2	$\log Y = -0.1137 + 0.6675 \log X$	0.694*	0.0418
	30	13	22.1	3.5	164.1	73.5			
	40	7	24.8	2.0	225.4	37.3			
	50	4	25.7	1.0	236.4	31.9			
15	20	7	18.4	1.5	83.3	22.6	$\log Y = 0.0926 + 0.6056 \log X$	0.908*	0.0120
	30	15	22.5	4.0	164.6	81.2			
	40	13	23.3	3.5	180.6	73.4			
	50	10	24.8	2.7	208.9	62.3			
20	30	15	19.7	3.7	130.9	75.5	$\log Y = 0.1936 + 0.6264 \log X$	0.865*	0.0224
	40	15	19.7	3.7	130.9	75.5			
	50	15	21.9	3.8	171.8	72.2			
	60	9	21.9	3.8	171.8	72.2			
25	30	6	25.0	1.0	219.9	29.5	$\log Y = -0.0594 + 0.7475 \log X$	0.941*	0.0288
	40	15	20.0	5.1	118.1	82.6			
	50	15	20.0	5.1	118.1	82.6			
	60	11	22.3	4.2	151.0	75.7			
30	30	7	25.5	1.3	207.5	35.7	$\log Y = 0.2000 + 0.6593 \log X$	0.790*	0.0270
	40	15	22.4	3.1	152.5	64.5			
	50	15	22.4	3.1	152.5	64.5			
	60	15	22.4	3.1	152.5	64.5			
34	30	6	24.1	1.8	180.3	36.9	$\log Y = 0.1338 + 0.7062 \log X$	0.548*	0.0430
	40	15	21.2	3.0	125.0	60.3			
	50	14	21.4	3.0	128.8	62.3			
	60	7	22.9	3.3	157.0	68.3			
		2	24.5	1.5	168.1	12.8			

* $P < 1\%$.

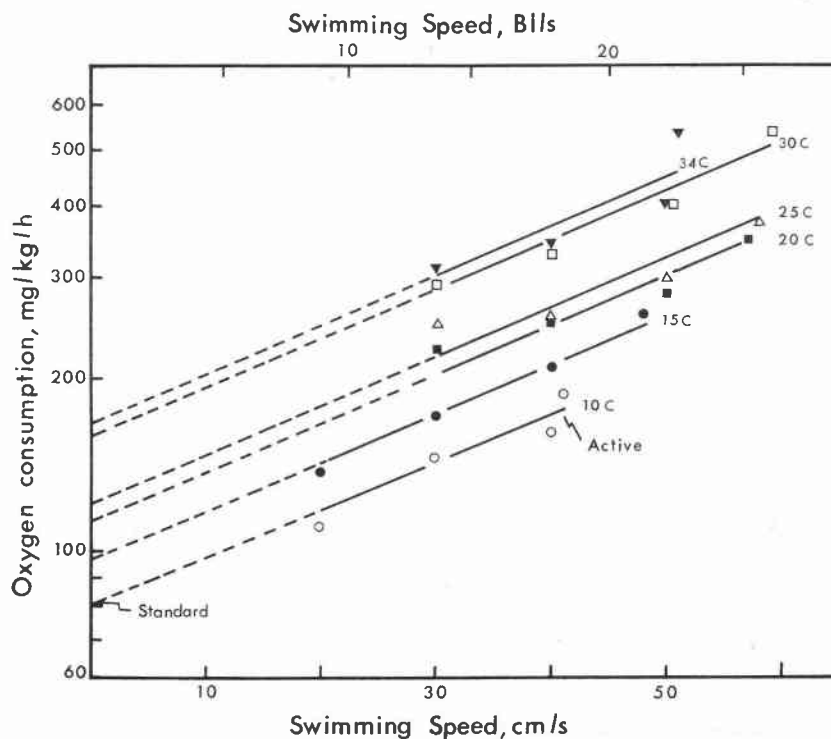


FIG. 3. The relation between oxygen consumption and swimming speed for largemouth bass of 150 g and a total length of 22.5 cm at different temperatures. Active represents oxygen consumption at the maximum sustained swimming speed.

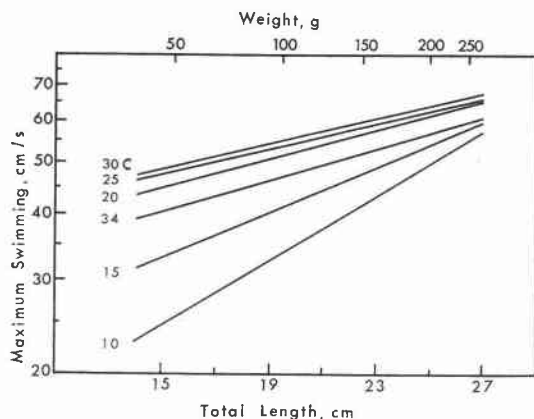


FIG. 4. Maximum sustained swimming speed of largemouth bass in relation to total length (weight) at different temperatures. Number of fish used, length, weight, regression equations, correlation coefficients, and confidence intervals of slopes are given for each temperature in appendix II available from the Depository of the National Research Council of Canada* in Ottawa, Canada.

*See previous footnote about obtaining copies of material from the Depository.

10 to 34 C when expressed on a logarithmic grid (Fig. 5). The Q_{10} for standard oxygen consumption was 1.32.

Maximum sustained swimming speed increased linearly with total length at each temperature when expressed on a semilogarithmic grid (Fig. 4). At 10 C, slope of the regression, 0.0303, was higher and significantly different ($P > 0.05$) than at other temperatures. The slope at 15, 0.0210, was less than that at 10 C but higher than that at other temperatures ($P > 0.05$). Significant differences among slopes were not demonstrable for temperatures from 20 to 34 C. For a given total length, within the range followed, maximum swimming speed increased from 10 to 30 C. Between 30 and 34 C, maximum swimming speed decreased.

Discussion

In respiration studies involving sustained swimming, training or conditioning of fish to a current of water is an important experimental procedure (Brett 1964; Farmer and Beamish

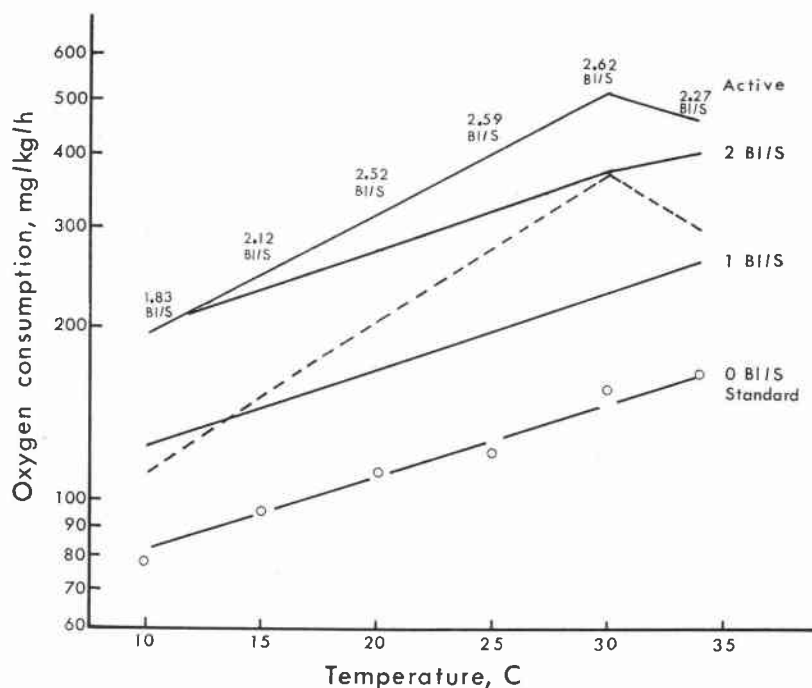


FIG. 5. Influence of temperature on oxygen consumption of a 150-g, 22.5-cm largemouth bass at different swimming speeds expressed as body lengths per second (BL/s). The metabolic scope for activity is denoted by a broken line.

1969; Kutty 1968; MacLeod 1967; Smit 1965). In any given trial oxygen consumption of largemouth bass decreased with time at the initial low swimming speed. Moreover, oxygen consumption for a given speed decreased and maximum sustained swimming speed increased over two or three trials and thereafter remained constant. In part, the decline in oxygen consumption with training may be attributed to a subsidence of excitability. Physiological changes such as an increase in haemoglobin and capacity of the heart with training (Hochachka 1961) may also relate to an increase in performance and a decrease in oxygen consumption. Hammond and Hickman (1966) found conditioned rainbow trout, *Salmo gairdneri*, were able to tolerate high concentrations of blood lactic acid during long-term exercise.

That oxygen consumption of largemouth bass was not altered when the concentration of oxygen was increased above air saturation suggests that the availability of oxygen at the lower concentration was sufficient to meet the metabolic expenditure without incurring a debt, at least over the swimming speeds followed at 25°C. Concordant

with observations on largemouth bass, active oxygen consumption of other species (e.g. goldfish) has been shown not to change with oxygen increase above a critical level (Basu 1959). It appears that each species has a critical oxygen threshold or incipient limiting level (Fry 1947) above which sufficient oxygen can be consumed to provide for the expenditure of energy associated with sustained swimming speeds.

Standard oxygen consumption is often estimated by extrapolation of the relation between locomotory activity, either spontaneous or forced, and oxygen consumption to zero activity. Provided the extrapolation is not long, the method presumably provides a reasonable estimate of maintenance requirements. The relation between standard oxygen consumption and temperature is generally described by one of three patterns. Perhaps the most common is a convex curve showing decreasing Q_{10} values with increasing temperature (Fry and Hart 1948; Beamish 1964a). A linear semilogarithmic increase in oxygen consumption with temperature as was found for largemouth bass has been found also for other species (Beamish 1964a).

Less frequent is a concave curve with Q_{10} increasing with temperature (Brett 1964). Common to all patterns is a continuous increase in maintenance requirement with temperature rise.

Few estimates have been made of standard or active oxygen consumption of largemouth bass. Glass (1968) determined routine oxygen consumption at 20 C in relation to deprivation of food. After the fish had fasted for 48 h, comparable to that followed in this study, oxygen consumption was 112 mg/kg h for a 33.6-g fish. At the same temperature, standard oxygen consumption in this study was also 112 mg/kg h but for a 150-g fish. Unless a direct relationship exists between weight and oxygen consumption or oxygen consumption was influenced by a difference in some environmental factor, such as photoperiod, estimates of standard oxygen consumption made in this study are probably high. MacLeod (1967) found an active oxygen consumption of 636 mg/kg h for a 1.6-g largemouth bass at 20 C. For a comparable temperature and weight active oxygen consumption was calculated as 830 mg/kg h in the present study.

Active oxygen consumption increased with temperature to 30 C (Fig. 5). That active oxygen consumption decreased between 30 and 34 C together with increased maintenance costs at the higher temperature may reflect limitations imposed by the reduction in ambient oxygen. In support of this suggestion, oxygen consumption increased linearly between 10 and 34 C for standard and the lower swimming speed. However, at the higher swimming speed (2 body lengths) with a rising demand for oxygen, coupled with a reduction of available oxygen at the upper temperatures, the linear increase extended only to 30 C. From 30 to 34 C the increase in oxygen consumption was only slight.

Energy available for swimming, termed metabolic scope for activity by Fry (1947), is calculated by subtraction of standard from active oxygen consumption (Fig. 5). For largemouth bass the scope increased with temperature to 30, and decreased from 30 to 34 C.

Customarily, swimming speed is expressed as cm/s or body lengths (BL/s). In general the observations on largemouth bass indicate a decrease in relative performance with size and are in accord with the earlier observations of Brett (1965) on sockeye salmon. Brett (1965) attributed this to an increase in hydrodynamic drag with

size which he suggests outweighs the advantages of increased metabolic scope and body musculature. In terms of absolute speed achieved there is, of course, a decided advantage with increase in size.

Proportionate changes in maximum sustained swimming speed in relation to temperature were independent of size from 20 to 34 C. However, at 10 and 15 C, slopes of the linear regressions were steeper than at the higher temperatures. Extrapolation of the relation above the size range followed suggests large fish should perform better at the lower temperatures. This seems unlikely. A more reasonable explanation is that estimates of maximum sustained swimming speeds at 10 and 15 C are low for small fish or that the size range was not sufficient to accurately determine slope values.

Few measurements of maximum sustained swimming speeds are available for largemouth bass. MacLeod (1967) determined short-term maximum speeds of 1.6-g largemouth bass at 20 C. On the assumption that total length was 5.7 cm, maximum speed was estimated to be 5.14 BL/s. In this study it was calculated that a fish of similar size could swim 5.85 BL/s, not far above the observed value of MacLeod.

Customarily, the influence of weight on oxygen consumption is described by a logarithmic linear regression. The logarithm of oxygen consumption for largemouth bass of different weight, swimming at the same absolute swimming speed, increased linearly with the logarithm weight; however, in each case the regression slope was considerably below unity. Similar observations have been made in other studies with other species (Rao 1968; Farmer and Beamish 1969; others). A trend of increasing slopes with increasing swimming speed found for *Tilapia nilotica* (Farmer and Beamish 1969) and rainbow trout, *Salmo gairdneri* (Rao 1968), was not apparent for largemouth bass. Rather, where comparisons were possible, slopes were about the same among the swimming speeds tested at a given temperature. In contrast, the slope of the relation between active oxygen consumption and weight approached unity. Concordant with observations of largemouth bass, Brett (1965) on sockeye salmon, Rao (1968) on rainbow trout, and Farmer and Beamish (1969) on *T. nilotica*, have all demonstrated a near direct weight relation for active oxygen consumption.

Acknowledgments

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- ALDER, H. L., and E. B. ROESSLER. 1964. Introduction to probability and statistics. 3rd ed. W. H. Freeman and Company, San Francisco and London.
- AMERICAN PUBLIC HEALTH ASSOCIATION. 1960. Standard methods for the examination of water, sewage and industrial wastes. 11th ed. Amer. Pub. Health Ass., New York. pp. 1-626.
- BASU, S. P. 1959. Active respiration of fish in relation to ambient concentrations of oxygen and carbon dioxide. J. Fish. Res. Board Can. **21**: 175-212.
- BEAMISH, F. W. H. 1964a. Respiration of fishes with special emphasis on standard oxygen consumption. II. Influence of weight and temperature on respiration of several species. Can. J. Zool. **42**: 177-188.
- . 1964b. Influence of starvation on standard and routine oxygen consumption. Trans. Amer. Fish. Soc. **93**: 103-107.
- BRETT, J. R. 1964. The respiratory metabolism and swimming performance of young sockeye salmon. J. Fish. Res. Board Can. **21**: 1183-1226.
- . 1965. The relation of size to rate of oxygen consumption and sustained swimming speed of sockeye salmon (*Oncorhynchus nerka*). J. Fish. Res. Board Can. **22**: 1491-1501.
- FARMER, G. J., and F. W. H. BEAMISH. 1969. Oxygen consumption of *Tilapia nilotica* in relation to swimming speed and salinity. J. Fish. Res. Board Can. **26**: 2807-2821.
- FRY, F. E. J. 1947. Effects of the environment on animal activity. Univ. Toronto Stud. Biol. Ser. 55 (Publ. Ontario Res. Lab. No. 68). pp. 5-62.
- . 1957. The aquatic respiration of fish. In The physiology of Fishes. Vol. I. Edited by M. E. Brown. Academic Press, New York. pp. 1-63.
- . 1967. Responses of vertebrate poikilotherms to temperature. In Thermal biology. Edited by A. H. Rose. Academic Press, London. pp. 375-409.
- FRY, F. E. J., and J. S. HART. 1948. The relation of temperature to oxygen consumption in the goldfish. Biol. Bull. **94**: 66-77.
- GLASS, N. R. 1968. The effect of time and food deprivation on the routine oxygen consumption of largemouth bass (*Micropterus salmoides*). Ecology, **49**: 340-343.
- HAMMOND, B. R., and C. P. HICKMAN, JR. 1966. The effect of physical conditioning on the metabolism of lactate, phosphate, and glucose in rainbow trout, *Salmo gairdneri*. J. Fish. Res. Board Can. **23**: 65-83.
- HOCHACHKA, P. W. 1961. Liver glycogen reserves of interacting resident and introduced trout populations. J. Fish. Res. Board Can. **18**: 125-135.
- KUTTY, M. N. 1968. Influence of ambient oxygen on the swimming performance of goldfish and rainbow trout. Can. J. Zool. **46**: 647-653.
- MACLEOD, J. C. 1967. A new apparatus for measuring maximum swimming speeds of small fish. J. Fish. Res. Board Can. **24**: 1241-1252.
- RAO, G. M. M. 1968. Oxygen consumption of rainbow trout (*Salmo gairdneri*) in relation to activity and salinity. Can. J. Zool. **46**: 781-786.
- SMIT, H. 1965. Some experiments on the oxygen consumption of goldfish (*Carassius auratus* L.) in relation to swimming speed. Can. J. Zool. **43**: 623-633.
- STEELE, G. D., and J. H. TORRIE. 1960. Principles and procedures of statistics. McGraw-Hill Book Company Inc., New York.